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(1) Research Program/Project/Study Title: Status of the Different Germplasm in CAPSU Coconut Breeding Trial Unit (CBTU)	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
Municipality/City:		Mambusao	
Province:		Capiz	
Region:		VI	
(5) Classification Research: ☐ Basic ☒ Applied Development: ☐ Pilot Testing ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity Coconut		(8) Discipline/Field of Specialization Agriculture	

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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

The study was conducted at the Coconut Breeding Trial Unit (CBTU) of CAPSU Burias, Mambusao, Capiz, Philippines, to evaluate the status of the different coconut germplasm. Data for the number of missing hills per variety, number of existing trees per variety, total number of hills per variety, and age of the trees were gathered.

Based on the evaluation, the area shows lack of maintenance and fertilization, leading to stunted and unhealthy growth of the existing trees. Visible yellowing and die back of leaves, and lack of fruiting were observed. Most hills of the different varieties planted in CBTU were missing, and some CBTU areas are still uncleaned. The area indicates the need for immediate and comprehensive rehabilitation to improve the health of the aging trees and promote production.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Pollen viability test, Pollen viability, coconut, coconut pollen, culture media

- 1. Introduction** (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Coconut (*Cocos nucifera* L.) is a staple crop in tropical regions, crucial for its economic, nutritional, and industrial applications. The Coconut Breeding Trial Unit (CBTU) in Burias, Mambusao, Capiz, Philippines, plays a pivotal role in the conservation and improvement of coconut germplasm. This unit is instrumental in the evaluation and development of superior coconut varieties, addressing both local and global demands for enhanced coconut production.

Germplasm conservation is critical for the long-term sustainability of coconut production, as it ensures the availability of genetic resources for breeding programs (Batugal & Engelmann, 1998). The CBTU houses a diverse collection of coconut germplasm, representing a wide range of genetic variations. This diversity is crucial for breeding programs aimed at developing new varieties with improved traits such as disease resistance, drought tolerance, and higher yield potential (Perera et al., 2009).

The status of germplasm in the CBTU involves evaluating the genetic diversity, health, and performance of different coconut accessions. Genetic diversity within a germplasm collection is vital for the adaptability and resilience of coconut varieties to changing environmental conditions and emerging pests and diseases (Santos et al., 1996). Studies have shown that maintaining a broad genetic base within coconut breeding programs can enhance the crop's ability to withstand biotic and abiotic stresses, thereby ensuring stable production (Arunachalam et al., 2003).

The CBTU utilizes various evaluation methods to assess the performance of different coconut germplasm. Morphological characterization, molecular markers, and field performance trials are commonly employed to identify superior accessions and understand their genetic relationships (Lebrun et al., 1998). For instance, molecular markers such as SSR (simple sequence repeat) and AFLP (amplified fragment length polymorphism) have been used to analyze the genetic diversity of coconut populations, providing valuable insights for breeding and conservation efforts (Perera et al., 2000).

Moreover, the CBTU focuses on the adaptability and performance of different coconut varieties under local conditions. The agro-climatic conditions in Burias, Mambusao, Capiz,

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including soil type, rainfall patterns, and temperature variations, significantly influence the growth and yield of coconut plants (FAO, 1995). Evaluating the performance of germplasm under these specific conditions helps identify varieties that are best suited for the region, ensuring higher productivity and sustainability.

Disease resistance is another critical aspect of germplasm evaluation at the CBTU. Coconut production in the Philippines is often threatened by diseases such as coconut cadang-cadang viroid (CCCVd) and bud rot, which can cause significant yield losses (Pacumbaba & Rajotte, 1994). By screening germplasm for resistance to these diseases, the CBTU can develop and promote varieties that are less susceptible to disease outbreaks, thereby protecting coconut yields and farmers' livelihoods.

In addition to disease resistance, the CBTU also evaluates germplasm for traits such as drought tolerance and oil content. Drought tolerance is increasingly important in the context of climate change, which is expected to exacerbate water scarcity in many coconut-growing regions (Magat, 1993). High oil content is a desirable trait for varieties used in copra and coconut oil production, which are significant economic drivers in the Philippines (Magat, 1999).

The integration of advanced breeding techniques, such as marker-assisted selection (MAS) and genomic selection, with traditional breeding methods can enhance the efficiency and effectiveness of germplasm evaluation and development at the CBTU (Santos et al., 1996). These technologies enable the identification of desirable traits at the genetic level, accelerating the breeding process and improving the precision of selection (Baudouin et al., 2006).

The status of different germplasm in the Coconut Breeding Trial Unit (CBTU) in Burias, Mambusao, Capiz, Philippines, is a crucial determinant of the success and sustainability of coconut breeding programs. By maintaining and evaluating a diverse collection of coconut germplasm, the CBTU contributes to the development of superior varieties that meet the needs of local and global markets.

Objectives of the Study

The study was conducted to evaluate the status of the different germplasm in CAPSU coconut breeding trial unit (CBTU). It specifically aimed to:

1. Determine the total number of coconut trees existed in CBTU;
2. Evaluate the total number of missing hills per variety; and
3. Provide the latest data of the different germplasm in CBTU.

(10) **Materials and Methods or Methodology** (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

MATERIALS AND METHODS

The study was conducted at the Coconut Breeding Trial Unit (CBTU) of CAPSU Burias, Mambusao, Capiz, Philippines.

Data collected included number of missing hills per variety, number of existing trees per variety, total number of hills per variety, and age of the trees.

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(11) **Results and Discussions** (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Age of the different coconut varieties in CBTU

The replicate of the PCA's coconut germplasm collection from the Zamboanga Research Center (ZRC) were planted in the Coconut Breeding Trial Unit (CBTU) in year 1989-1999. Tall varieties such as Aguinaldo, Bago Oshiro*, Bago Oshiro, Baybay, Gatusan, Gazelle, Karkar, Laguna*, Laguna, Macapuno, Markham Valley, Rennel Island, Salambuyan, San Ramon, Spicata, Tagnanan, Tahiti, Tampakan, West African, Vanuatu, and Zamboanga, and dwarf varieties like Baguer, Banga, Cameron Red, Galas, Kapatagan, Kinabalan, La Victoria Brown, Magtuod, Makilala, Malayan Yellow, Pilipog, San Isidro, Santo Niño, Sri Lanka, Tacunan, and Talisay were planted in year 1993.

Moreover, 28 years old Aromatic, New Buswang, Catigan, Equatorial Guinea, La Victoria Green, Malayan Red, and Tupi dwarf varieties, and hybrids varieties: MRD x TAC, MRD x BAY, MRD x WAT, MRD x CAT, MAT x MYD, MAT x CRD, and Agta tall were planted in 1995. Mañgipod was planted in 1996, Banigan, Culaman, Loong and Venus in 1997, and TAGT x CATD (ERAP) hybrid variety in 1999.

Lastly, hybrid varieties such as MRD x LAG, MYD x WAT, TACD x WAT, CAT x LAG were planted in late 1989.

Current number of missing and existing hills

The data in **Figure 1a, Figure 1b, Figure 1c, and Figure 1d** showed the current number of missing and existing hills of the tall coconut varieties in CBTU. As of July 2024, a total of 1,048 trees were currently existed and 1,440 are missing and will be replanted, including the accessions. Out of the total number hills counted, 52%-89% missing hills were recorded in Agta (26), Aguinaldo (28), Bago Oshiro (29), Banigan (42), Culaman (31), Gatusan (30), Gazelle (28), Karkar (32), Loong (22), Markham Valley (29), Rennel Island (28), Salambuyan, Spicata (37), West African (44), Venus (29), and Zamboanga (38). The highest number of existing trees was counted in the varieties such as Bago Oshiro S1 (40), Baybay (32), Laguna S1 (36), Laguna (37), Macapuno (35), Tagnanan (34), Tahiti (28), Tampakan (31), and Vanuatu (36), while the San Ramon has 50 hills counted with a 50% existing and 50% missing hills.

Additionally, the dwarf coconuts have a total of 634 plant stand and 930 missing hills. As shown in **Figures 2a, 2b, 2c, and 2d**, varieties such as Baguer, Galas, Kapatagan, La Victoria Green, Makilala, San Isidro, Santo Niño, Sri Lanka, and Talisay recorded a higher number of trees existed than the missing hills with the counts of 24, 32, 26, 26, 39, 40, 38, 17, and 28, respectively. The highest missing over the total number of hills was counted in Tupi with 37 missing hills and 3 trees existed. While the average missing hills of Aromatic (127), Banga (28), New Buswang (33), Catigan (117), Cameron (21), Equatorial Guinea (65), Kinabalan (29), la Victoria Brown (37), Mañgipod (25), Malayan Yellow (40), Malayan red (55), Pilipog (43), and Tacunan (64) ranges from 51%- 78% as of July 2024. Additionally, the Magtuod dwarf has 25 trees existed and 25 missing hills.

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Finally, **Figures 3a** and **3b** shows the missing and existing hills of the hybrid varieties. Most of the trees planted were missing in all the varieties. A total of 321 existing and 786 missing hills were counted in the hybrid varieties.

Clearing and underbrushing of the CBTU area

It was noticeable that the area was poorly maintained, with seed nuts and seedlings struggling to grow amidst overgrown grass and trees. The area clearly indicates an urgent need for rehabilitation. **Figures 4a** and **4b** shows the photo taken before and after the clearing and underbrushing activities conducted in CBTU. However, 25% of the are still uncleaned and cannot covered by the project's budget.

Status of the existing trees

The lack of maintenance and fertilization has led stunted and unhealthy growth of coconuts, which severely impact the health and overall productivity of the area. The age of the trees ranged from 25-35 years old, yet some have failed to produce fruits. This lack of fruiting, combined with the visible yellowing and die back of leaves, suggests that trees may be suffering from nutrient deficiencies or other stress factors. This situation underscores the need for immediate and comprehensive rehabilitation of the overall area to improve the health of the aging trees and promote production.

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate hedings if they are long. Justify paragraphs with main headings in caps.)

Conclusions

Based on the evaluation of the area, the following recommendations are drawn:

1. Most hills of the different varieties planted in CBTU were missing.
2. The area shows lack of maintenance and fertilization, leading to stunted and unhealthy growth of the existing trees.
3. Lack of fruiting, visible yellowing and die back of leaves were also observed.
4. Some CBTU areas are still uncleaned and cannot covered by the project's budget.

Recommendations

Based on the evaluation of the area, the following recommendations are forwarded:

1. Urgent rehabilitation of the Coconut Breeding Trial Unit is needed.
2. Fertilization of the existing trees to improve health and overall productivity of the area.
3. Maintenance of the area by conducting clearing, underbrushing of the area and cleaning of the trees, especially the area that were still uncleaned.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

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(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.)

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APPENDICES

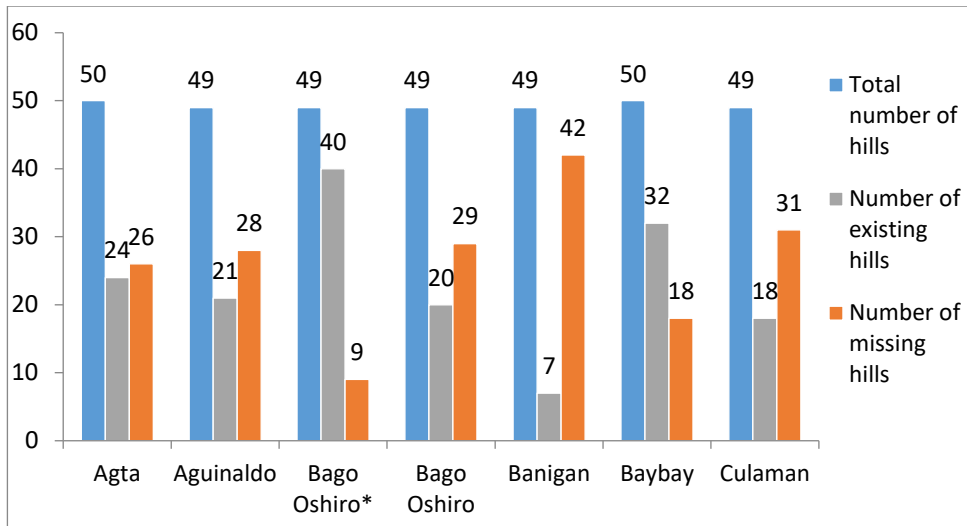


Figure 1a. Number of missing and existing number of hills of the different tall coconut variety/ies as of July 2024.

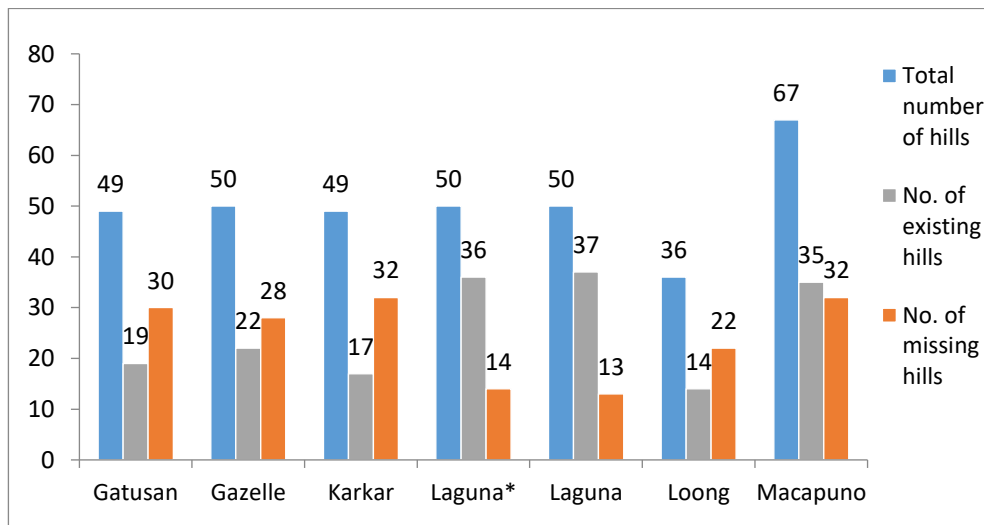


Figure 1b. Number of missing and existing number of hills of the different tall coconut variety/ies as of July 2024.

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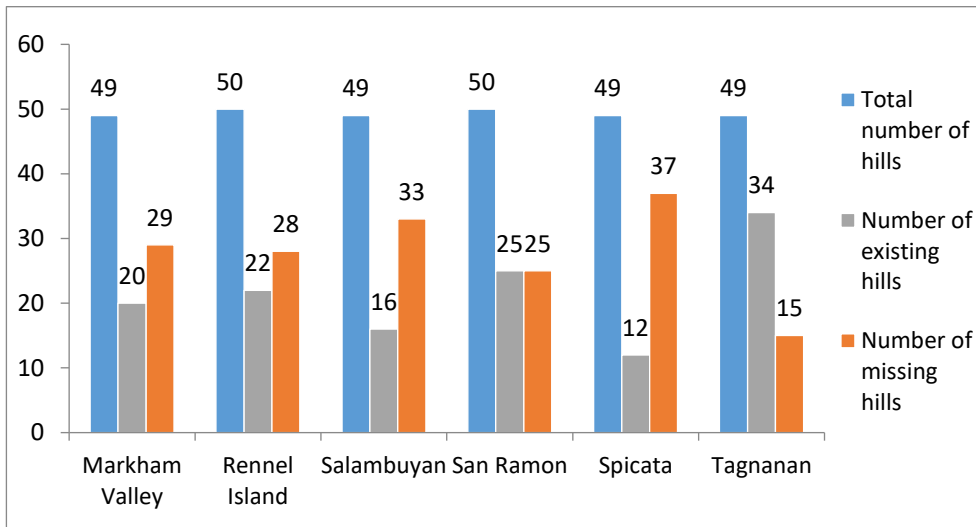


Figure 1c. Number of missing and existing number of hills of the different tall coconut variety/ies as of July 2024.

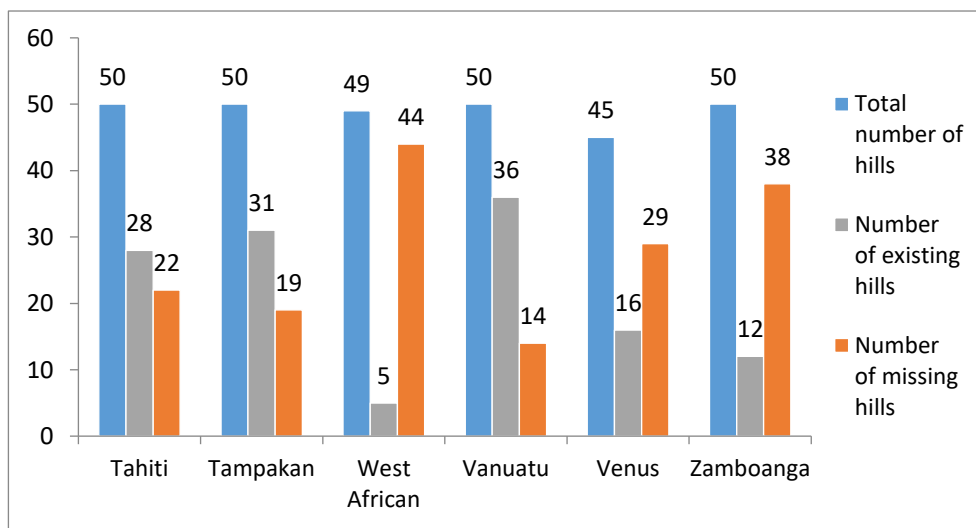


Figure 1d. Number of missing and existing number of hills of the different tall coconut variety/es as of July 2024.

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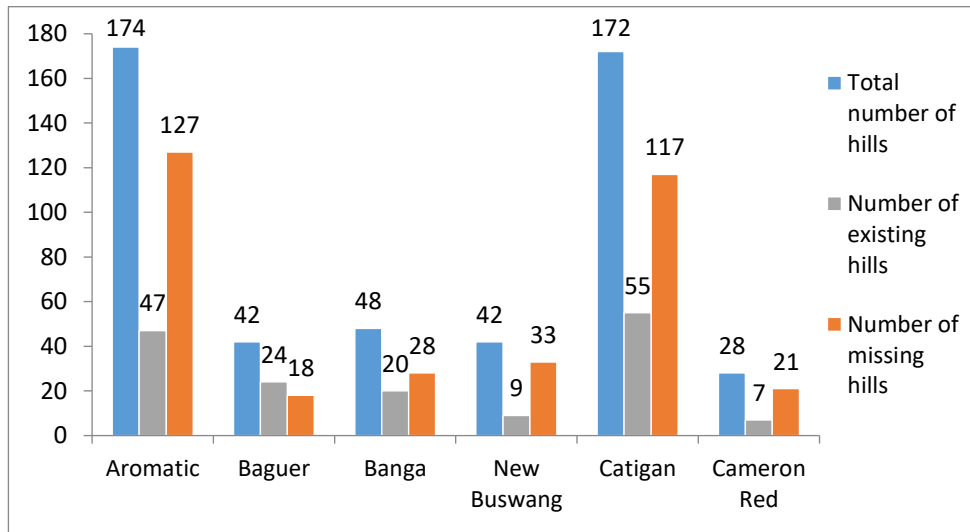


Figure 2a. Number of missing and existing number of hills of the different dwarf coconut variety/ies as of July 2024.

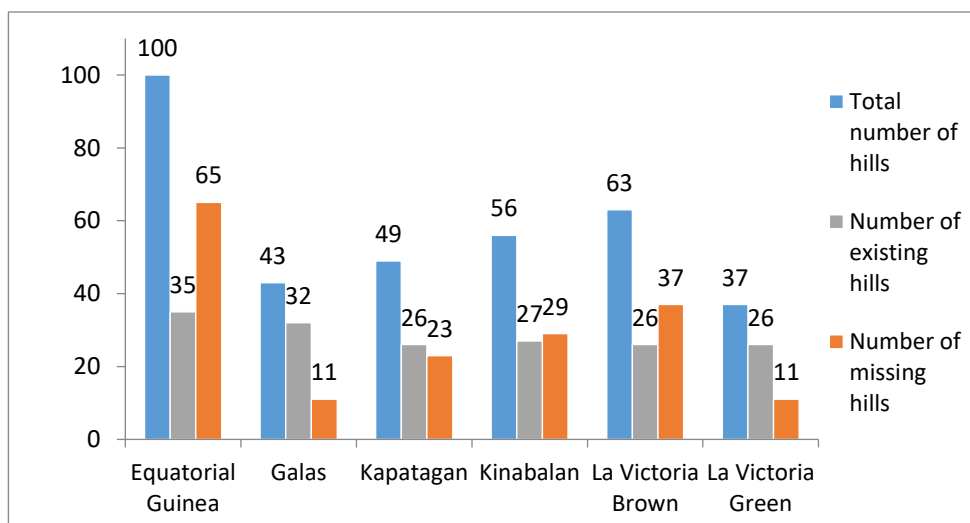


Figure 2b. Number of missing and existing number of hills of the different dwarf coconut variety/es as of July 2024.

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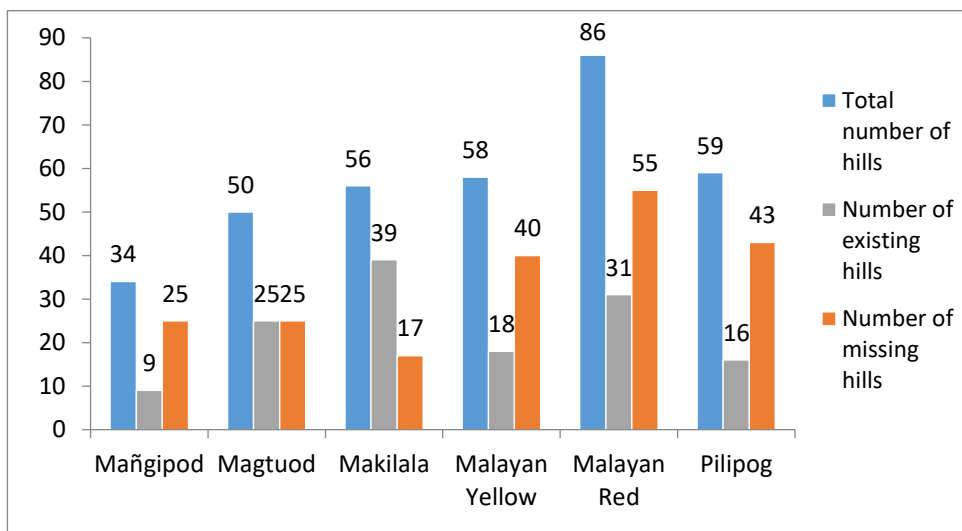


Figure 2c. Number of missing and existing number of hills of the different dwarf coconut variety/ies as of July 2024.

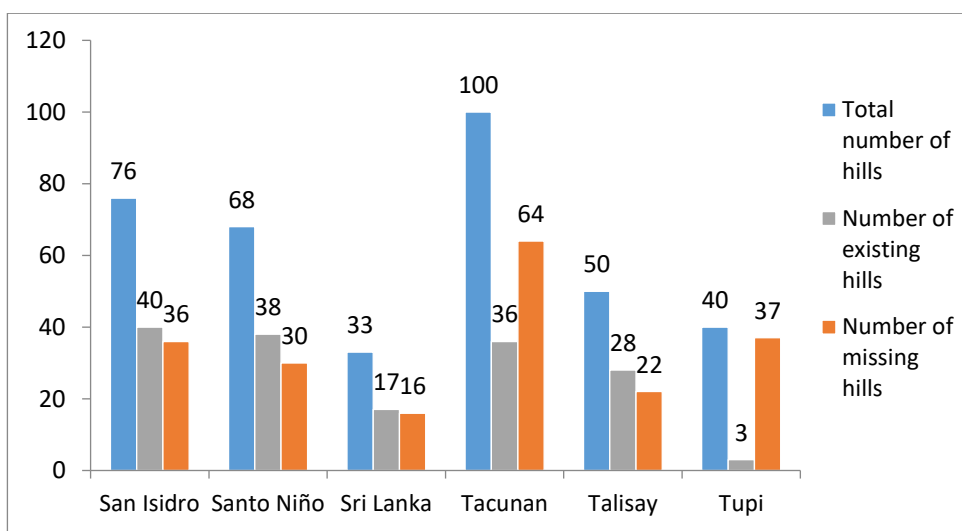


Figure 2d. Number of missing and existing number of hills of the different dwarf coconut variety/ies as of July 2024.

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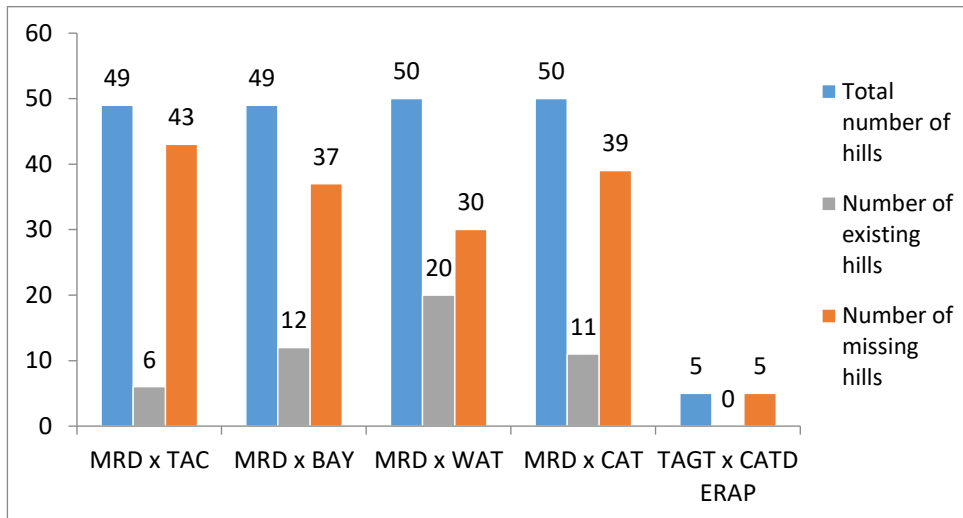


Figure 3a. Number of missing and existing number of hills of the different hybrid coconut variety/ies as of July 2024.

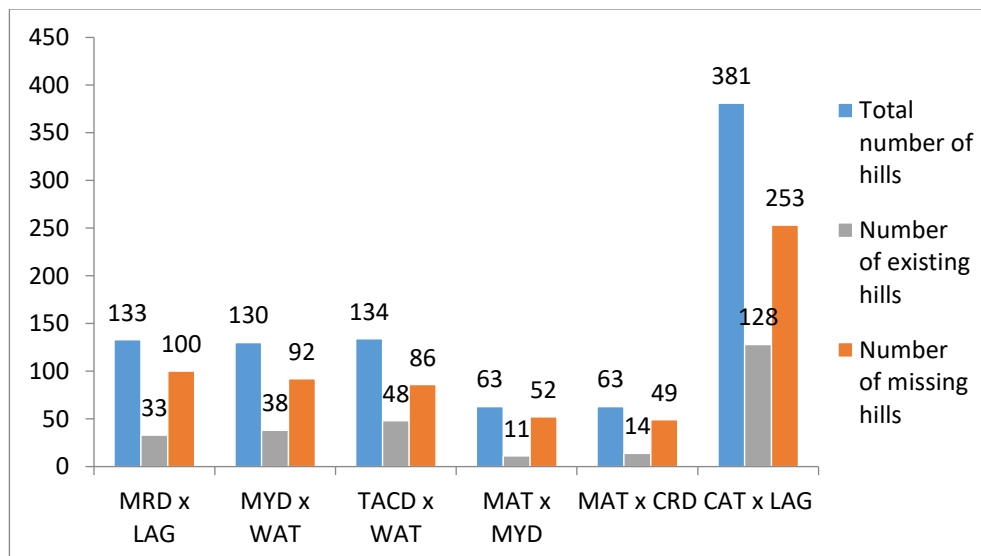


Figure 3b. Number of missing and existing number of hills of the different hybrid coconut variety/ies as of July 2024.

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Figure 4a. Before the clearing and underbrushing of CBTU.



Figure 4b. After the clearing and underbrushing of CBTU.

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Table 1. Status of the coconut germplasm in CBTU as of July 2024

Name	Code	Origin	Date Planted	Hills			Area (ha)	Age (years)
				Total	Existin g	Missin g		
Accessions								
1. Acc. #16 North Cotabato Tall	NOC T	Zamboanga Research Center (ZRC)	November 1997	50	16	34	0.35	26
2. Acc. #17 Pikit Tall	PIKT	Zamboanga Research Center (ZRC)	May 1997	49	17	32	0.34	27
3. Acc. #18 Pangasinan Tall	PANT	Zamboanga Research Center (ZRC)	October 1995	49	18	31	0.34	28
4. Acc. #19 Santiago Island Tall	SANT	Zamboanga Research Center (ZRC)	October 1995	50	20	30	0.35	28
5. Acc. #20 La Union Tall	LAN T	Zamboanga Research Center (ZRC)	October 1995	50	22	28	0.35	28
6. Acc. #21 Ilocos Sur Tall	ILOT	Zamboanga Research Center (ZRC)	October 1995	49	37	12	0.34	28
7. Acc. #23 Cotabato Tall	COTT	Zamboanga Research Center (ZRC)	November 1997	54	11	43	0.38	26
8. Acc. #25 Spring Taluya Tall	SPRT	Zamboanga Research Center (ZRC)	May/Nove mber 1997	50	16	34	0.35	27
9. Acc. #26 Dalican Tall	DAL T	Zamboanga Research Center (ZRC)	October 1995	50	19	31	0.35	28
10. Acc. #27	BALT	Zamboanga Research	October 1993	49	11	38	0.34	30

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Ballesteros Tall		Center (ZRC)						
11. Acc. #28 Cagayan Tall	CAG T	Zamboanga Research Center (ZRC)	October 1995	39	0	39	0.27	28
12. Acc. #29 Sanchez Mira Tall	SCHT	Zamboanga Research Center (ZRC)	October 1995	42	11	31	0.29	28
13. Acc. #30 Botolan Tall	BOTT	Zamboanga Research Center (ZRC)	October 1995	40	26	14	0.28	28
14. Acc. #31 Samal Island Tall	SAIT	Zamboanga Research Center (ZRC)	October 1995	46	16	30	0.32	28
15. Acc. #32 Bukidnon Tall	BUK T	Zamboanga Research Center (ZRC)	October 1995	50	18	32	0.35	28
16. Acc. #33 Bataan Tall	BAA T	Zamboanga Research Center (ZRC)	November 1997	46	21	25	0.32	26
17. Acc. #34 Zambales Tall	ZABT	Zamboanga Research Center (ZRC)	October 1995	49	15	34	0.34	28
18. Acc. #35 Namukon Tall	NAM T	Zamboanga Research Center (ZRC)	October 1995	50	23	27	0.35	28
19. Acc. #36 Bani Tall	BAN T	Zamboanga Research Center (ZRC)	October 1995	49	15	34	0.34	28
20. Acc. #37 Pugo Tall	PUGT	Zamboanga Research Center (ZRC)	October 1995	47	27	20	0.33	28
21. Acc. #40 Upi Tall	UPIT	Zamboanga Research Center (ZRC)	May 1997	47	18	29	0.33	27

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22. Acc. #42 Leon Tall	LEOT	Zamboanga Research Center (ZRC)	November 1997	49	22	27	0.34	26
23. Acc. #44 Antique Tall	ANT T	Zamboanga Research Center (ZRC)	October 1996	45	14	31	0.32	27
24. Acc. #46 Anda Tall	AND T	Zamboanga Research Center (ZRC)	November 1997	54	23	31	0.38	26
25. Acc. #47 Tigbaon Tall	TIGT	Zamboanga Research Center (ZRC)	November 1997	49	15	34	0.34	26
Talls								
1. Agta Tall	AGA T	Bago Oshiro, Davao City	October 1995	50	24	26	0.35	28
2. Aguinaldo Tall	AGD T	Agui, Naga, Zamboanga City	September 1993	49	21	28	0.34	30
3. Bago Oshiro Tall S1	BAO T*	Bago Oshiro, Davao City	September 1993	49	40	9	0.34	30
4. Bago Oshiro Tall	BAO T	Bago Oshiro, Davao City	September 1993	49	20	29	0.34	30
5. Banigan Tall	BNG T	Banigan, Liloy, Zamboanga del Norte	May 1997	49	7	42	0.34	27
6. Baybay Tall	BAY T	Baybay, Leyte	September 1993	50	32	18	0.35	30
7. Culaman Tall	CULT	Culaman, Malita, Davao del Sur	May 1997	49	18	31	0.34	27
8. Gatusan Tall	GAT T	Davao City	September 1993	49	19	30	0.34	30
9. Gazelle Peninsula Tall	GPT	Papua New Guinea	September 1993	50	22	28	0.35	30
10. Karkar Tall	KKT	Papua New Guinea	September 1993	49	17	32	0.34	30

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11. Laguna Tall S1	LAG T*	Davao City	September 1993	50	36	14	0.35	30
12. Laguna Tall	LAG T	Davao City	September 1993	50	37	13	0.35	30
13. Loong Tall	LON T	Loong, Concepcion , Iloilo	October 1996/ November 1997	36	14	22	0.25	26/27
14. Macapuno 23B	MAC T	Bago Oshiro, Davao City	October 1995	67	35	32	0.47	28
15. Markham Valley Tall	MVT	Papua New Guinea	September 1993	49	20	29	0.34	30
16. Rennel Island Tall	RIT	Solomon Island	September 1993	50	22	28	0.35	30
17. Salambuyan Tall	SALT	Salambuyan , Lapuyao, Zamboanga del Sur	September 1993	49	14	35	0.34	30
18. San Ramon Tall	SNRT	Zamboanga City	September 1993	50	25	25	0.35	30
19. Spicata Tall	SPIT	Davao City	September 1993	49	12	37	0.34	30
20. Tagnanan Tall	TAG T	Mabini, Davao del Norte	September 1993	49	34	15	0.34	30
21. Tahiti Tall	PYTT	IRHO, Ivory Coast	September 1993	50	28	22	0.35	30
22. Tampakan Tall	TPKT	Tampakan, South Cotabato	September 1993	50	31	19	0.35	30
23. Vanuatu Tall	VTT	IRHO, Ivory Coast	September 1993	50	36	14	0.35	30
24. Venus Tall	VEN T	Venus, Osmeña, Zamboanga del Norte	May 1997	45	16	29	0.32	27
25. West African Tall	WAT	IRHO, Ivory Coast	September 1993	49	5	44	0.34	30
26. Zamboanga Tall	ZAM T	Zamboanga City	September 1993	50	12	38	0.35	30
Dwarfs								

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1. Aromatic Green Dwarf	ARO D	Thailand	October 1993/1995	174	47	127	0.96	28/30
2. Baguer Green Dwarf	BAG D	Libungan, North Cotabato	October 1993	42	24	18	0.23	30
3. Banga Green Dwarf	BAN D	Banga, South Cotabato	October 1993	48	20	28	0.27	30
4. New Buswang Green Dwarf	BUS D	New Buswang, Kalibo, Aklan	October 1995	42	9	33	0.23	28
5. Catigan Green Dwarf	CAT D	Bago Oshiro, Davao City	June 1989/October 1995	172	55	117	0.95	28/35
6. Cameron Red Dwarf	CRD	IRHO, Ivory Coast	October 1993	28	7	21	0.16	30
7. Equatorial Guinea Green Dwarf	EGD	IRHO, Ivory Coast	October 1993/1995	100	35	65	0.55	28/30
8. Galas Green Dwarf	GAL D	Galas, Dipolog City	October 1993	43	32	11	0.24	30
9. Kapatagan Green Dwarf	KAP D	Kapatagan, Lanao Norte	October 1993	49	26	23	0.27	30
10. Kinabalan Green Dwarf	KIND	Malita, Davao del Sur	October 1993	56	27	29	0.31	30
11. La Victoria Brown Dwarf	VIBD	Aurora, Zamboanga del Sur	October 1993	63	26	37	0.35	30
12. La Victoria Green Dwarf	VICD	Aurora, Zamboanga del Sur	October 1993	37	26	11	0.21	28
13. Mañgipod Green Dwarf	MÑD D	Zamboanga Research Center (ZRC)	October 1996	34	9	25	0.19	27

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14. Magtuod Green Dwarf	MAG D	Magtuod, Davao City	October 1993	50	25	25	0.28	30
15. Makilala Green Dwarf	MAK D	Makilala, North Cotabato	October 1993	56	39	17	0.31	30
16. Malayan Yellow Dwarf	MYD	Hijo, Davao City	October 1993	58	18	40	0.32	30
17. Malayan Red Dwarf	MRD	Hijo, Davao City	October 1993/1995	86	31	55	0.48	28/30
18. Pilipog Green Dwarf	PILD	Davao City	October 1993	59	16	43	0.33	30
19. San Isidro Green Dwarf	SNID	San Isidro, Agusan del Sur	October 1993	76	40	36	0.42	30
20. Santo Niño Green Dwarf	SÑO D	Santo Niño, Davao Norte	October 1993	68	38	30	0.38	30
21. Sri Lanka Green Dwarf	SGD	IRHO, Ivory Coast	October 1993	33	17	16	0.18	30
22. Tacunan Green Dwarf	TAC D	Tacunan, Davao City	October 1993	100	36	64	0.55	30
23. Talisay Green Dwarf	TAL D	Talisay, Dapitan City	October 1993	50	28	22	0.28	30
24. Tupi Green Dwarf	TUPD	Tupi, South Cotabato	October 1995	40	3	37	0.22	28
Hybrids								
1. MRD x TACD		Zamboanga Research Center (ZRC)	October 1995	49	6	43	0.34	28
2. MRD x BAYT	PCA 15-3	Zamboanga Research	October 1995	49	12	37	0.34	28

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		Center (ZRC)						
3. MRD x WAT		Zamboanga Research Center (ZRC)	October 1995	50	20	30	0.35	28
4. MATD x MYD		Zamboanga Research Center (ZRC)	October 1995	63	11	52	0.44	28
5. MATD x CRD		Zamboanga Research Center (ZRC)	October 1995	63	14	49	0.44	28
6. CATD x LAGT	PCA 15-1	Zamboanga Research Center (ZRC)	June 1989	381	128	253	2.67	35
7. MRD x LAG	PCA 15-13	Zamboanga Research Center (ZRC)	June 1989	133	33	100	0.93	35
8. MYD x WAT		Zamboanga Research Center (ZRC)	June 1989	130	38	92	0.91	35
9. TACD x WAT	PCA 15-12	Zamboanga Research Center (ZRC)	June 1989	134	48	86	0.94	35
10. MRD x CATD		Zamboanga Research Center (ZRC)	October 1995	50	11	39	0.35	28
11. TAGT x CATD	ERAP	Zamboanga Research Center (ZRC)	May 1999	5	0	5	0.04	25